Corrections (red). Wilson lines are real orthogonal, $U^{\dagger\, bc} = U^{cb}$. The unmeasured gluon at z created after the shock carries $U^{\dagger bc}(z) = U^{cb}(z)$. A gluon created before the shock carries $U^{ce}(y)\rho^e$. A rotated source is $U^{ce}(x)\rho^e(x)$. Check: every amplitude bracket vanishes for a uniform target $U(x) \equiv U_0$.

Group - II crossection derivation

1 Group II

1.1 Row I [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\left[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{ac'}(w') - \bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{c'b}(z)\right]\times$$
$$\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

We write it in two parts: [Corrected: duplicated sentence removed.]

1.1.1 First part: [Done ✓]

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^4}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\frac{1}{[p^+(x-z)^2+k^+(x-w)^2]}$$
$$\left[\frac{\delta^{ij}\left[(x-z)^2-(x-w)^2\right]}{2\,(p^++k^+)\,(z-w)^2}+\frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right]\times\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^4}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\frac{1}{[p^+(x-z)^2+k^+(x-w')^2]}$$
$$\left[\frac{\delta^{ij}\left[(x-z)^2-(x-w')^2\right]}{2\,(p^++k^+)\,(z-w')^2}+\frac{1}{p^+}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w')^i}{(z-w')^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j\,(x-w')^i}{2\,(x-w')^2}\right]\right]\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{1.1}$$

Integrating w.r.t. p^+

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ & \\
-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times & \\
\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] & \\
+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} & \\
-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\} & \\
\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] & \\
+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ & \\
\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w')^2}{\Lambda\,(x-z)^2+k^+\,(x-w')^2}\right]\times & \\
\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right] & \\
-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2} & \\
+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\} & \\
\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] &
\end{aligned}
\tag{1.2}
$$

1.1.2 Second part: [Done ✓]

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\frac{p^+\,k^+}{(p^++k^+)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\,\frac{1}{[p^+\,(x'-z)^2+k^+\,(x'-w')^2]} & \\
\Bigg\{\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]+\frac{(p^++k^+)}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}\right.\right. & \\
\left.\left.+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\left[\frac{\delta^{ij}\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w')^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x'-z)^j\,(z-w')^i}{(z-w')^2}-\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-w')^2}\right]\right. & \\
\left.+\frac{(p^++k^+)}{p^+}\,\left[\frac{(x'-w')^i\,(z-w')^j}{(z-w')^2}+\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-z)^2}\right]\right]\Bigg\}\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{c'b}(z)U^{d'e'}(x')\,\rho^{e'}(x')\right] & \\
\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] &
\end{aligned}
\tag{1.3}
$$

Reducing the Wilson lines,

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\frac{p^+\,k^+}{(p^++k^+)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\,\frac{1}{[p^+\,(x'-z)^2+k^+\,(x'-w')^2]}$$
$$\left\{\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]+\frac{(p^++k^+)}{p^+}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}\right.\right.\right.$$
$$\left.\left.\left.+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\left[\frac{\delta^{ij}\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w')^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x'-z)^j\,(z-w')^i}{(z-w')^2}-\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-w')^2}\right]\right.$$
$$\left.+\frac{(p^++k^+)}{p^+}\,\left[\frac{(x'-w')^i\,(z-w')^j}{(z-w')^2}+\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-z)^2}\right]\right]\right\}\times\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)\right.$$
$$-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^d(x)$$
$$\left.+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{1.4}$$

Simplifying, we get

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\frac{p^+\,k^+}{(p^++k^+)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\,\frac{1}{[p^+\,(x'-z)^2+k^+\,(x'-w')^2]}$$
$$\left\{\frac{d_\perp\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(z-w)^2\,(z-w')^2}\right.$$
$$+\frac{(p^++k^+)}{k^+}\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]$$
$$+\frac{(p^++k^+)}{p^+}\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]$$
$$+\frac{(p^++k^+)^2}{k^{+\,2}}\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$\left.-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]$$
$$+\frac{(p^++k^+)^2}{p^{+\,2}}\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]$$
$$+\frac{(p^++k^+)^2}{p^+k^+}\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$\left.\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\right\}\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{1.5}$$

Integrating w.r.t. p^+ and we write it in 6 different parts:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\left\{\frac{1}{[(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2]}\times\right.$$
$$\left[\frac{(x-w)\cdot(z-w')(x'-z)\cdot(z-w)}{(z-w)^2(z-w')^2} - \frac{(x-w)\cdot(x'-w')(z-w)\cdot(x'-z)}{2(z-w)^2(x'-w')^2}\right.$$
$$+\frac{(x-z)\cdot(x'-z)(x-w)\cdot(z-w')}{2(x-z)^2(z-w')^2} - \frac{(x'-z)\cdot(x-z)(x'-w')\cdot(x-w)}{4(x-z)^2(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)(x-z)\cdot(z-w')}{(z-w)^2(z-w')^2} + \frac{(x'-w')\cdot(z-w)(x-z)\cdot(x'-z)}{2(z-w)^2(x'-z)^2}$$
$$\left.-\frac{(x-z)\cdot(z-w')(x'-w')\cdot(x-w)}{2(x-w)^2(z-w')^2} - \frac{(x'-w')\cdot(x-w)(x'-z)\cdot(x-z)}{4(x-w)^2(x'-z)^2}\right]\times$$
$$\left[\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+(\vee-k^+)(x'-z)^2}\right] - \log\left[\frac{\Lambda(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right]\right]\Bigg\}\times$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x) + N_c\,U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right] \tag{1.6}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{1}{[(x'-w')^2(x-z)^2(x-w)^2-(x-w)^4(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2} + \frac{(x'-w')\cdot(x-w)(x'-z)\cdot(z-w)}{2(z-w)^2(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')(x-z)\cdot(z-w')}{2(x-z)^2(z-w')^2} + \frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-z)^2(x'-z)^2}\right]\times\left[(x-z)^2 - \frac{(x-w)^2(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x) + N_c\,U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{(x-z)^2}{[(x'-w')^2(x-z)^2(x-w)^2-(x-w)^4(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2} + \frac{(x'-w')\cdot(x-w)(x'-z)\cdot(z-w)}{2(z-w)^2(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')(x-z)\cdot(z-w')}{2(x-z)^2(z-w')^2} + \frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-z)^2(x'-z)^2}\right]\times\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}\right]\times$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x) + N_c\,U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-z)^2}{[(x'-w')^4(x-z)^2-(x-w)^2(x'-z)^2(x'-w')^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2} + \frac{(x'-w')\cdot(x-w)(x'-z)\cdot(z-w)}{2(z-w)^2(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')(x-z)\cdot(z-w')}{2(x-z)^2(z-w')^2} + \frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-z)^2(x'-z)^2}\right]\times\log\left[\frac{(\vee-k^+)(x'-z)^2+k^+(x'-w')^2}{\Lambda(x'-z)^2+k^+(x'-w')^2}\right]$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x) + N_c\,U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right] \tag{1.7}$$

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{1.8}
$$

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{k^+ + \Lambda}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]} \\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
&\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
&\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times \\
&\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
&\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times \\
&\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]\times \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{1.9}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{k^+\,(x'-w')^2+(\vee-k^+)\,(x'-z)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{1.10}
$$

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\log\left[\frac{\vee}{\Lambda+k^+}\right]\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}$$
$$\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{c'b}(z)U^{d'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^{e}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}$$
$$\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$
$$+\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\left[\frac{1}{\vee}-\frac{1}{k^++\Lambda}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$
$$+\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}$$
$$\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{1.11}$$

1.2 Row II [Done ✓]

Only 4 ρ's

1.3 Row III [Done ✓]

Only 4 ρ's

1.4 Row IV [Done ✓]

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y,y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[\bar{\mathbb{C}}^{dac}_{1jik}(p^++k^+,y-w;p^+,w-z)$$
$$U^{db}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_k(p^++k^+,-y)-U^{db}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^++k^+,y-w;p^+,w-z)\bar{\mathbb{A}}^{(1)\,c}_k(p^++k^+,-y)\Big]$$

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w',x'} e^{-ik(w'-w)}\int_{x,y,y'}\frac{(w-z)^m\,(x-y)^k\,(w'-z)^{m'}\,(x'-y')^{k'}}{(w-z)^2\,(x-y)^2\,(w'-z)^2\,(x'-y')^2}\,\frac{p^+\,k^+}{(p^++k^+)^4}$$
$$\delta^{(2)}\left[y-w+\frac{p^+}{(p^++k^+)}\,(w-z)\right]\times\delta^{(2)}\left[y'-w'+\frac{p^+}{(p^++k^+)}\,(w'-z)\right]\times$$
$$\left[\delta_{km}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right]\times\left[\delta_{k'm'}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]$$
$$\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x)-f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right] \tag{1.12}$$

Integrating w and w',

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)\,\frac{(k^+ + p^+)}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^+ + p^+)^2} + \frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'} + \frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^e(x) - f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^e(x)\right] \tag{1.13}$$

Reducing Wilson lines, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)}\,e^{-ik\,(y'-y)\,\frac{p^+}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^+ + p^+)^2} + \frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'} + \frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.14}$$

We are performing z integral and using $\overline{MS}$ and $d_\perp = 2-\epsilon$, we get

$$\int_z\frac{(y-z)^m\,(y'-z)^{m'}}{(y-z)^2\,(y'-z)^2} = -\frac{\pi}{2}\left[\frac{2}{\epsilon} + 2\gamma_E + \log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\delta^{mm'} - \pi\,\frac{(y'-y)^m\,(y'-y)^{m'}}{(y'-y)^2} \tag{1.15}$$

This gives us

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^+ + k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[\frac{p^+}{(k^+ + p^+)^2} + \frac{p^+}{k^{+\,2}} + \frac{1}{p^+}\right]\times\left[\frac{1}{\epsilon} + \gamma_E + \frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]$$
$$\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^+ + k^+}{k^+}}\,\frac{(x-y)^k\,(x'-y')^{k'}\,(y'-y)^m\,(y'-y)^{m'}}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^+ + p^+)^2} + \frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'} + \frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.16}$$

Simplifying, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^+ + k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{p^+}{(k^+ + p^+)^2} + \frac{p^+}{k^{+\,2}} + \frac{1}{p^+}\right]\times$$
$$\left[\frac{1}{\epsilon} + \gamma_E + \frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^+ + k^+}{k^+}}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,\frac{d_\perp\,p^+}{(k^+ + p^+)^2}\right.$$
$$\left.+\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{p^+}{k^{+\,2}} + \frac{1}{p^+}\right]\right]\times\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.17}$$

Using $\xi = \frac{k^+}{(p^+ + k^+)}$, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+ + \Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\xi\,(1-\xi) + \frac{(1-\xi)}{\xi} + \frac{\xi}{(1-\xi)}\right]\frac{1}{\xi^2}$$
$$\left[\frac{1}{\epsilon} + \gamma_E + \frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+ + \Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\xi^2}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi(1-\xi)\right.$$
$$\left.+\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{(1-\xi)}{\xi} + \frac{\xi}{(1-\xi)}\right]\right]\times\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.18}$$

We write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{8\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ +\Lambda}}\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Bigg\{\mathcal{P}_{gg}(\xi)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[d_\perp\left[-\frac{1}{\epsilon}-\gamma_E-\frac{1}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]-1\right]+2\,N_c\,\xi(1-\xi)\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\right.$$
$$\left.-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\Bigg\}\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.19}$$

The pole is present in the following:

$$\frac{d^3N_{\rm pole}}{d^3k} = -\,\frac{g^2}{8\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{d_\perp}{k^+}\,\frac{1}{\epsilon}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ +\Lambda}}\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\mathcal{P}_{gg}(\xi)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.20}$$

where $\mathcal{P}_{gg}(\xi) = 2\,N_c\,\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]$ is the gluon - gluon splitting function. Since, we have handled the UV divergence, now we can safely use plus prescription to integrate wrt ξ.

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]$$
$$\left[\int_{\frac{k^+}{\vee}}^{1}\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left[\frac{(\vee-k^+)\,(k^++\Lambda)}{\vee\,\Lambda}\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\,\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ +\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.21}$$